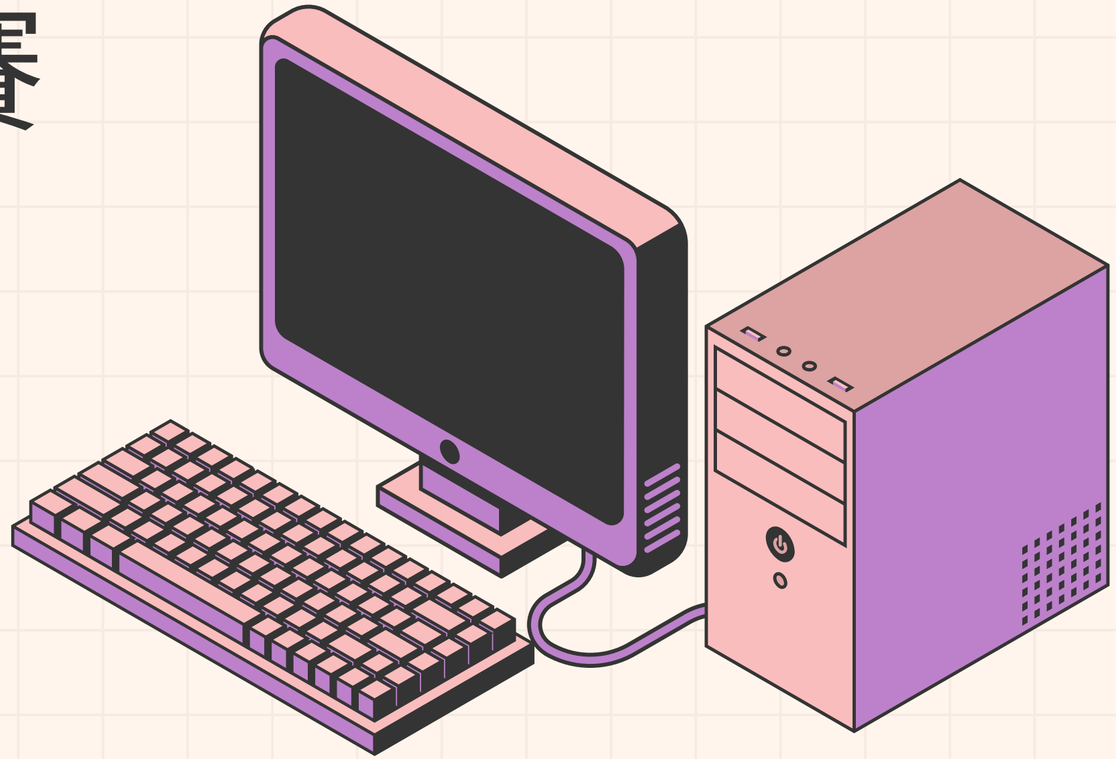
GROUP 3

HPC-I 2026 模擬比賽



HPL 數據

	N	NB	P*Q	time	Gflops	state
1	36000	128	2*4	>36分	?	fail
2	24000	128	2*4	121.14	76.08	pass
3	30000	256	2*4	177.78	101.25	pass
4	26000	256	1*2	33.71	347.61	pass

數據1 --> 數據2：擴大 N (76.08 --> 101.25 GFLOPS)

在 RAM 足夠的前提下，N 設的越大，CPU 計算時間
就越長，網路傳輸延遲的時間佔比就越低，最終測
出來的 GFLOPS 就會越高

如果 N 設太大超過 RAM 的容量，電腦就會卡死

數據2 --> 數據3：加大 BLOCK (101.25 --> 123.44 GFLOPS)

CPU 快取的速度比 RAM 快上百倍。如果 NB 設得剛剛好，就能完美塞滿 CPU 的 L2/L3 快取，減少cache miss

NB 太小：COMPULSORY CACHE MISS 暴增
SPATIAL LOCALITY 失效
NB 太大：CACHE THRASHING

數據3 --> 數據4： (123.44 --> 347.61 GFLOPS)

1.將 P×Q 從 2×4 改成 1×2
2.編譯時加入 -march=native 開啟 AVX2 向量指令集。
3.將矩陣反向縮小至 N=26000。

STUDIO SHODWE

ICON

1. DEPENDENCIES

Compiled all libraries from source: ZLIB → HDF5 → NetCDF-C/Fortran → OpenBLAS → FYAML → XML2

Used a single custom prefix so nothing conflicted with system libraries

Ran git lfs install before cloning — without this, all binary files are silently broken stubs

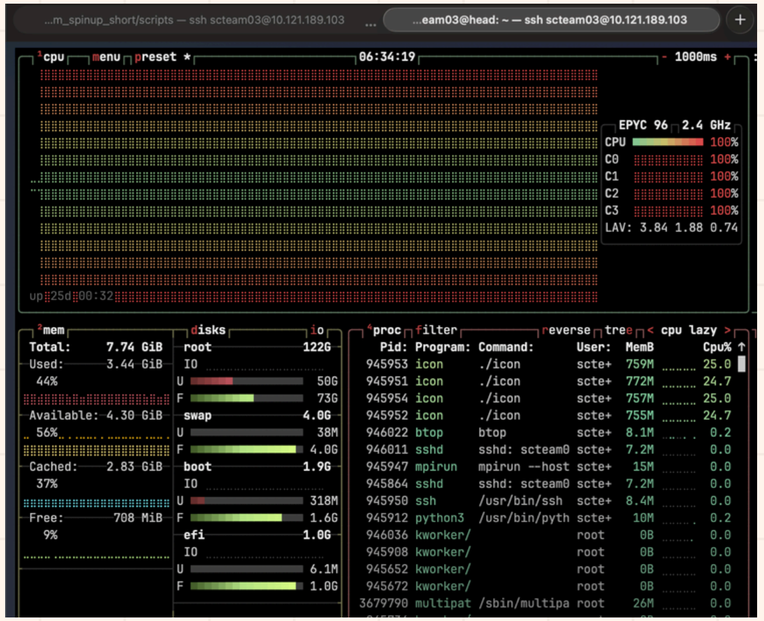
Built ICON with -march=native for AVX2 SIMD on our CPUs

ICON

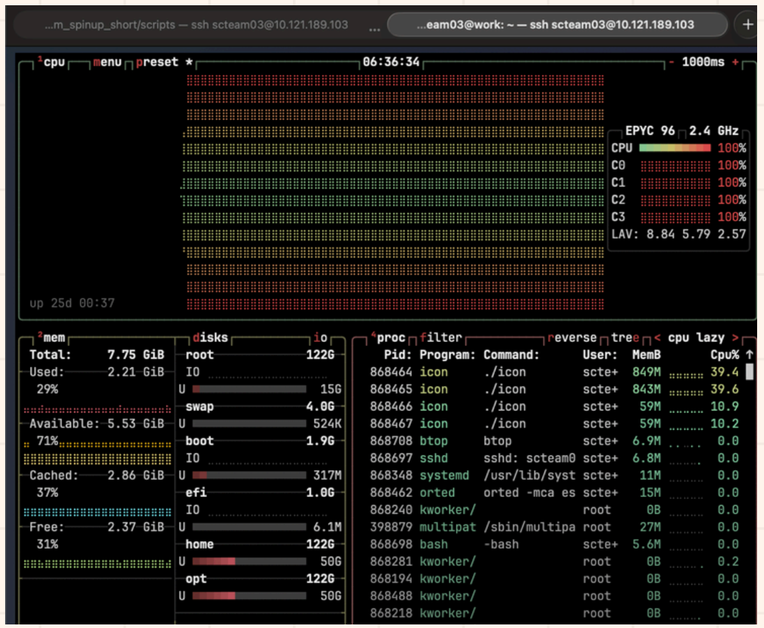
2. RUN SCRIPT MODIFICATIONS

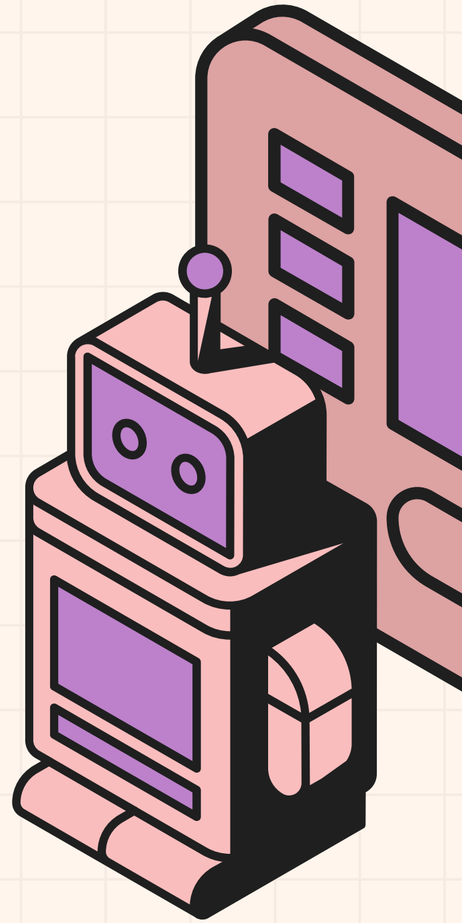
- Original script was written for Slurm — our cluster has none
- Stubbed out scontrol() and set SLURM_JOBID=999999 to prevent crashes
- Replaced Slurm MPI launch with explicit: mpirun --host head:4,work:4 -n 8
- ICON initialized cleanly — restart file loaded, grid decomposed across 8 ranks

*Head node



*Work node

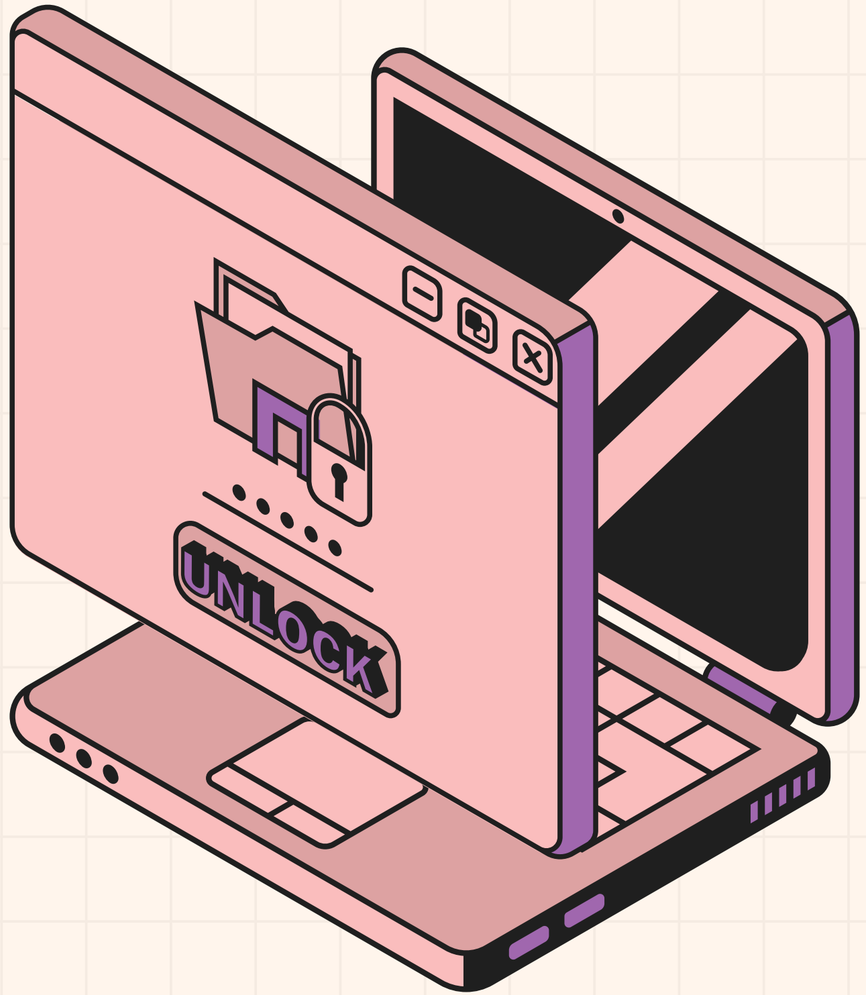




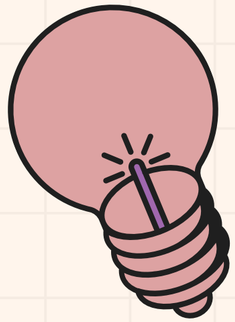
ICON

3. PROBLEMS

- Simulation started and timesteps were advancing — but wall-clock time ran out before the output interval was reached
- Root cause of time loss: two multi-node MPI bugs that were hard to diagnose
- Head node has two network interfaces → MPI registered on one, sent on the other → silent packet drops
- Fix: force TCP traffic to internal subnet only (OMPI_MCA_btl_tcp_if_include=10.21.21.0/24)
- ICON uses MPI one-sided (RMA) operations → osc/sm component broke when sm BTL was disabled → MPI_ERR_WIN on init
- Fix: OMPI_MCA_osc=pt2pt
- These two issues alone cost several hours before the actual simulation could even start

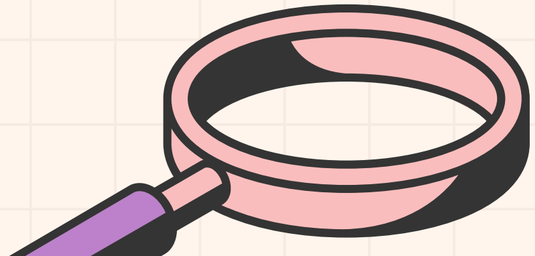


ICON



4. WHAT WE'D FIX TO GET OUTPUT

- Increase dynamics timestep from PT1M → PT4M (4× fewer solver calls)
- nproma=32 already set for AVX2 vectorization
- Separate I/O procs (num_io_procs=2) so disk writes don't block compute



MLPerf Llama2-7B

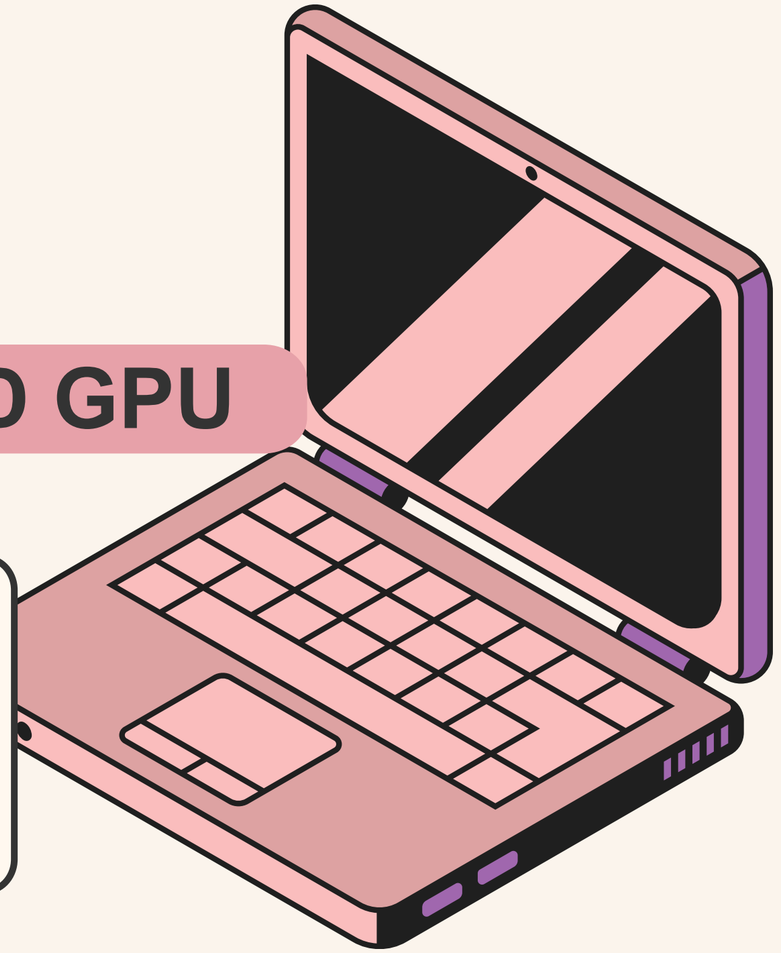
From “it won’t run”
to 4.54x throughput

Offline · PerformanceOnly · single AMD GPU

262.6
baseline
tokens/s

1193.3
final tokens/

4.54x
gain



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This was full-stack benchmark execution, not just one config file.

Any broken layer makes the tokens/s number meaningless.

Crashes, stale configs, or multi-GPU spillover would invalidate the story.

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I kept the official benchmark constraints fixed.

Fixed

Llama2-7B

Offline

PerformanceOnly

1 AMD GPU

Changed

Batching

Concurrency

GPU memory cap

vLLM switches

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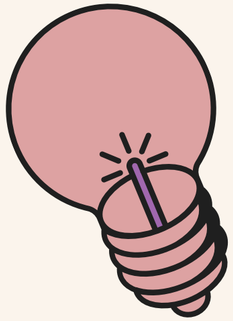
Most taxing blocker: storage exhaustion.



No container meant no benchmark, no baseline, and no tuning.

most taxing

100%	99%	315GB
/home full	/ root full	/dev/shm free

Observed failure: FATAL no space left on device

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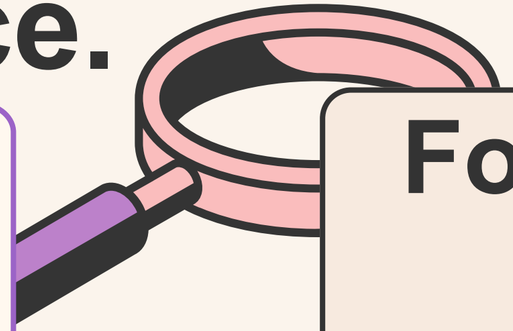
The workaround was to turn RAM into build scratch space.

Redirect volatile build pressure

APPTAINER_TMPDIR
APPTAINER_CACHEDIR
/dev/shm/...



Force hardcoded paths

symlink hidden .apptainer_build paths into RAM

Result: an 8.0GB persistent .sif image was generated.

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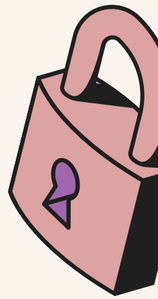
The hidden blocker was configuration shadowing.



The job ran, but my YAML edits were invisible inside Apptainer.

Host edits
local YAML

→

Not mounted
container
ignores

→

Stale default
0.70 mem cap

This created a false sense of tuning until the mount was fixed.

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The fix was a bind mount plus a sanity check.

--bind host worksp

/home/.../closed/AMD

→

/lab-mlperf-inference

Read YAML inside

cat from container before Slurm run

After this, every tuning result became trustworthy.

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The valid baseline exposed underutilization.

The default vLLM configuration was safe, not fast.

262.59	1.102	0.70	8
tokens/s	samples/	mem cap	seqs

Hypothesis: Offline throughput needs more active sequences per GPU pass.

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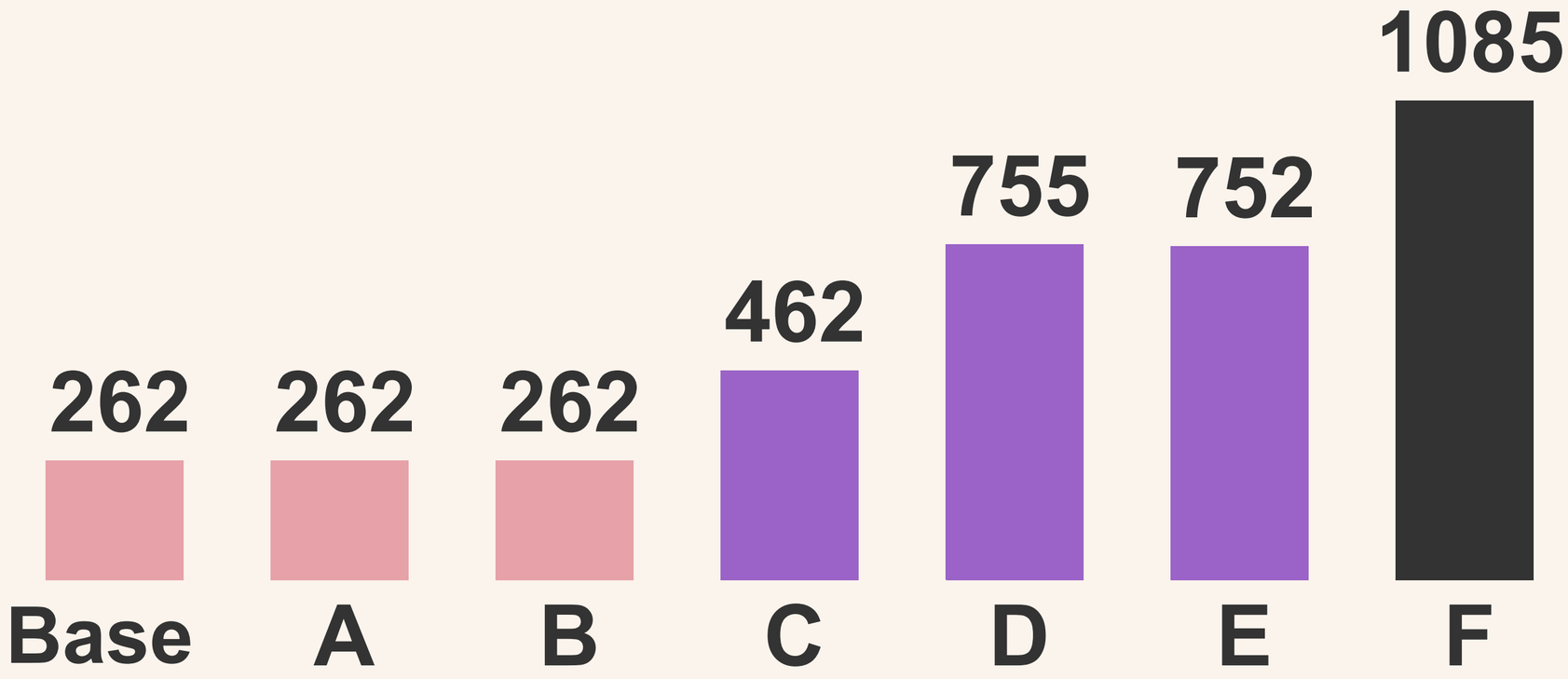
Sequence concurrency was the winning knob.

Token batch size alone barely moved throughput.



Run F

16384 tokens
64 seqs
1084.520 t/s

A/B stayed flat; C/D/F jumped when max_num_seqs increased.

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Extra vLLM tweaks were tested one at a time.

Chunked prefill won after the Run F batching setup.

1084.5	1088.3	1193.3	1115.1
F batch setup	eager off	chunked prefill	scheduler=2

Important: one-knob tests, not combined results.

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Final valid result: 4.54x more tokens/s.

Same model, same dataset, same Offline single-GPU rule.

262.591	→	1193.320	4.54x
baseline		final	gain

Final config: 0.90 mem · 16384 tokens · 64 seqs · chunked prefill

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Appendix: Base to B confirm token window alone did not help.

Base	A	B
2048 tok · 8 seq · 0.70	4096 tok · 8 seq · 0.70	8192 tok · 8 seq · 0.70
262.591 t/s	**262.011 t/s**	**262.330 t/s**

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Appendix: C to F show sequence concurrency scaling.

C	D	E	F
8192 tok	8192 tok	16384 tok	16384 tok
16 seq · 0.80	32 seq · 0.80	32 seq · 0.90	64 seq · 0.90
462.105 t/s	**755.355 t/s**	**752.100 t/s**	**1084.520 t/s**

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Appendix: G to I were one-knob tests after Run F.

G	H	I
enforce_eager False	chunked prefill True	scheduler steps = 2
1088.250 t/s	**1193.320 t/s**	**1115.070 t/s**

NOT combined.

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HEMELB
OPTIMIZATIONS

杜武明光 - 113006214

THE IMPORTANCE OF SPACK

COMPILE LIBRARY WITHOUT PRIVILAGES

We can’t install dependencies with Ubuntu’s package manager, which requires root privileges. Spack allows us to install dependencies without

EASE OF INSTALLATION

Spack compile the library automatically, which can also be easily configured.

CONTAINERIZED ENVIRONMENT

Can create environments to create different build with different dependencies, compilers and features without affect other builds.

HEMELB
COMPILATION

CMake arguments

- -DHEMELB_GPU_BACKEND=HIP_ROCM
- -DCMAKE_CXX_COMPILER=hipcc
- -DAMDGPU_TARGETS="gfx1201"
- -DCMAKE_HIP_FLAGS="-O3 -fopenmp --offload-arch=gfx1201 -funsafe-math-optimizations -ffast-math -Rpass-analysis=kernel-resource-usage"
- -DHEMELB_USE_PARMETIS=ON
- -DCMAKE_HIP_ARCHITECTURES=gfx1201
- -DHEMELB_USE_SSE3=ON

HEMELB
OPTIMIZATION - REGISTER PRESSURE

```
/cuda_params.cu:1757:0: Function Name: _ZN6hemelb71GPU_CollideStream_mMidFluidCollision_mWallCollision_sBB_WallShearStress
/cuda_params.cu:1757:0:     TotalSGPRs: 91
/cuda_params.cu:1757:0:     VGPRs: 148
/cuda_params.cu:1757:0:     ScratchSize [bytes/lane]: 320
/cuda_params.cu:1757:0:     Dynamic Stack: False
/cuda_params.cu:1757:0:     Occupancy [waves/SIMD]: 9
/cuda_params.cu:1757:0:     SGPRs Spill: 0
/cuda_params.cu:1757:0:     VGPRs Spill: 0
/cuda_params.cu:1757:0:     LDS Size [bytes/block]: 0
```

Reduced VGPR down to around 50, but performance worsens.

OPTIMIZATION - INPUT FILE

- Logging to files require moving data from GPU back to CPU, costing time.
- After removing non-necessary logging (only log what the problem ask for), performance improved.
- Increasing logging period worsens performance.

杜武明光 - 113006214

VISUALIZATIONS

- Compile hemeXtract
- Run: hemeXtract -X whole.dat -o file.txt
- Import the file into Paraview

杜武明光 - 113006214

ParaView 6.1.1

File Edit View Sources Filters Extractors Tools Catalyst Macros Help

Time: 0 0

Representation

Pipeline Browser

- builtin:
- ANUplane3.txt
- TableToPoints1
- Calculator1
- Glyph1

Properties Information

Properties

Apply Reset Delete Help

Search ... (use Esc to clear text)

Properties ('Glyph1' / Glyph)

Glyph Source

Glyph Type: Arrow

Orientation

Orientation Array: velocity

Scale

Scale Array: No scale array

Scale Factor: 0.000635

Masking

Glyph Mode: All Points

Display (SpreadSheetRepresentatio

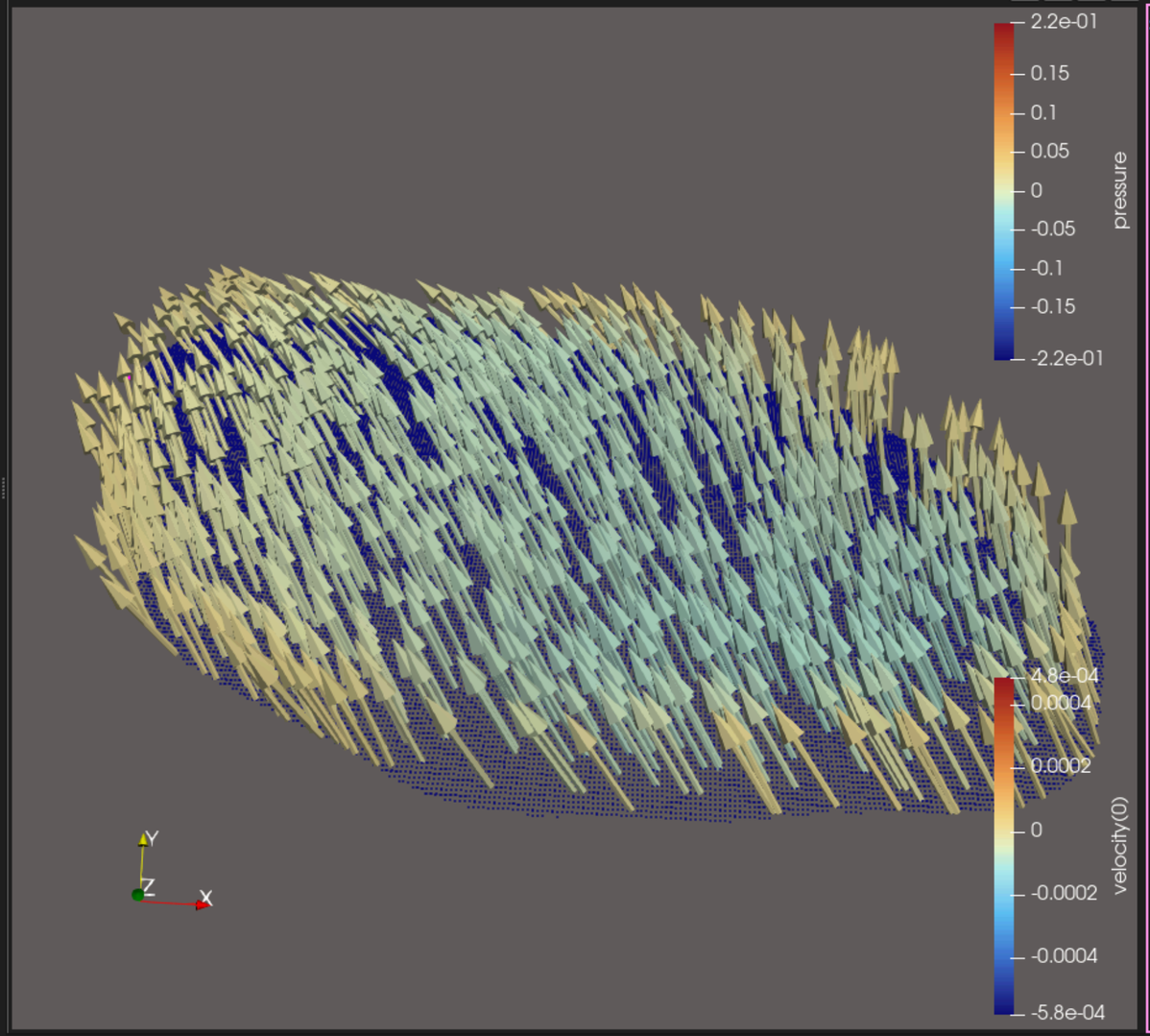
View (SpreadSheet View)

Cell Font Size: 9

Header Font Size: 9

Layout #1

RenderView1



SpreadSheetView1

Showing: Glyph1 Attribute: Point Data Precision: 6

	Point ID	Points			Points_Magnitude	pressure	step	
0	0	0.0109869	0.0410312	0.0222686	0.04796	-0.191686	5000	-4.
1	1	0.0111334	0.0406716	0.0224087	0.0477523	-0.191686	5000	-4.
2	2	0.0110008	0.0410404	0.0222778	0.0479753	-0.191686	5000	-4.
3	3	0.0111473	0.0406809	0.0224179	0.0477678	-0.191686	5000	-4.
4	4	0.0110174	0.0410438	0.0222691	0.047978	-0.191686	5000	-4.
5	5	0.0111639	0.0406842	0.0224092	0.0477704	-0.191686	5000	-4.
6	6	0.0110201	0.0410379	0.0222512	0.0479652	-0.191686	5000	-4.
7	7	0.0111666	0.0406784	0.0223913	0.0477576	-0.191686	5000	-4.
8	8	0.0110062	0.0410286	0.022242	0.0479499	-0.191686	5000	-4.
9	9	0.0111527	0.0406691	0.0223821	0.0477422	-0.191686	5000	-4.
10	10	0.0109896	0.0410253	0.0222507	0.0479473	-0.191686	5000	-4.
11	11	0.0111361	0.0406658	0.0223908	0.0477396	-0.191686	5000	-4.
12	12	0.0109869	0.0410312	0.0222686	0.04796	-0.191686	5000	-4.
13	13	0.0110008	0.0410404	0.0222778	0.0479753	-0.191686	5000	-4.
14	14	0.0110174	0.0410438	0.0222691	0.047978	-0.191686	5000	-4.
15	15	0.0110201	0.0410379	0.0222512	0.0479652	-0.191686	5000	-4.
16	16	0.0110062	0.0410286	0.022242	0.0479499	-0.191686	5000	-4.
17	17	0.0109896	0.0410253	0.0222507	0.0479473	-0.191686	5000	-4.
18	18	0.0111361	0.0406658	0.0223908	0.0477396	-0.191686	5000	-4.
19	19	0.0111527	0.0406691	0.0223821	0.0477422	-0.191686	5000	-4.
20	20	0.0111666	0.0406784	0.0223913	0.0477576	-0.191686	5000	-4.
21	21	0.0111639	0.0406842	0.0224092	0.0477704	-0.191686	5000	-4.
22	22	0.0111473	0.0406809	0.0224179	0.0477678	-0.191686	5000	-4.
23	23	0.0111334	0.0406716	0.0224087	0.0477523	-0.191686	5000	-4.
24	24	0.0109246	0.0412281	0.0221845	0.0480755	-0.191686	5000	-4.

Bazket: 17.5 GiB/31.7 GiB 55.4%

ParaView 6.1.1

File Edit View Sources Filters Extractors Tools Catalyst Macros Help

Time: 0 0

velocity(1) Surface

Pipeline Browser

builtin:
cow-output.txt
TableToPoints1

Properties Information

Properties

Apply Reset Delete Help

Search ... (use Esc to clear text)

Properties ('TableToPoints1

X Column grid_x
Y Column grid_y
Z Column grid_z
2D Points
Keep All Data Arrays

Display (GeometryRepresei

Representation Surface

Coloring

velocity(1)

Edit

Styling

Opacity 1

Lighting

Specular 0

Layout #1

RenderView1

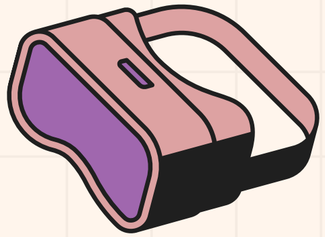
1.2e-38
1e-38
8e-39
6e-39
4e-39
2e-39
0.0e+00
velocity(1)

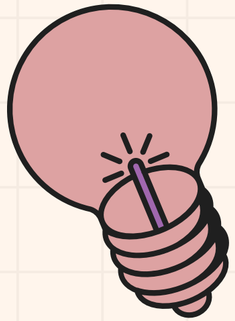
SpreadSheetView1

Showing TableToPoints1 Attribute: Point Data Precision: 6

	Point ID		Points		Points_Magnitude	pressure	step	velocity(0)	velocity(1)	velocity(2)
12627202	12627...	0.001...	0.003104	0.001005	0.00377254	6.54617e-18	20000	0	0	0
12627203	12627...	0.001...	0.003104	0.001011	0.00377415	6.54617e-18	20000	0	0	0
12627204	12627...	0.001...	0.003104	0.001018	0.00377603	6.54617e-18	20000	0	0	0
12627205	12627...	0.001...	0.00311	0.000973	0.00376909	6.54617e-18	20000	0	0	0
12627206	12627...	0.001...	0.00311	0.000979	0.00377065	6.54617e-18	20000	0	0	0
12627207	12627...	0.001...	0.00311	0.000986	0.00377247	6.54617e-18	20000	0	0	0
12627208	12627...	0.001...	0.00311	0.000992	0.00377404	6.54617e-18	20000	0	0	0
12627209	12627...	0.001...	0.00311	0.000998	0.00377562	6.54617e-18	20000	0	0	0
12627210	12627...	0.001...	0.00311	0.001005	0.00377748	6.54617e-18	20000	0	0	0
12627211	12627...	0.001...	0.00311	0.001011	0.00377908	6.54617e-18	20000	0	0	0
12627212	12627...	0.001...	0.00311	0.001018	0.00378096	6.54617e-18	20000	0	0	0
12627213	12627...	0.001...	0.003117	0.000973	0.00377487	6.54617e-18	20000	0	0	0
12627214	12627...	0.001...	0.003117	0.000979	0.00377642	6.54617e-18	20000	0	0	0
12627215	12627...	0.001...	0.003117	0.000986	0.00377824	6.54617e-18	20000	0	0	0
12627216	12627...	0.001...	0.003117	0.000992	0.00377981	6.54617e-18	20000	0	0	0
12627217	12627...	0.001...	0.003117	0.000998	0.00378139	6.54617e-18	20000	0	0	0
12627218	12627...	0.001...	0.003117	0.001005	0.00378325	6.54617e-18	20000	0	0	0
12627219	12627...	0.001...	0.003117	0.001011	0.00378484	6.54617e-18	20000	0	0	0
12627220	12627...	0.001...	0.003117	0.001018	0.00378672	6.54617e-18	20000	0	0	0
12627221	12627...	0.001...	0.003072	0.000973	0.00374135	6.54617e-18	20000	0	0	0
12627222	12627...	0.001...	0.003072	0.000979	0.00374292	6.54617e-18	20000	0	0	0
12627223	12627...	0.001...	0.003072	0.000986	0.00374475	6.54617e-18	20000	0	0	0
12627224	12627...	0.001...	0.003072	0.000992	0.00374634	6.54617e-18	20000	0	0	0
12627225	12627...	0.001...	0.003072	0.000998	0.00374793	6.54617e-18	20000	0	0	0
12627226	12627...	0.001...	0.003072	0.001005	0.0037498	6.54617e-18	20000	0	0	0
12627227	12627...	0.001...	0.003072	0.001011	0.00375141	6.54617e-18	20000	0	0	0

Bazket: 19.9 GiB/31.7 GiB 62.8%





THANK YOU